

Appendix

The technical appendix covers the following aspects:

- A. Experimental Details
- B. Hyperparameter Sensitivity Analysis
- C. Discussion and Future Work

A. Experimental Details

Dataset Our experiments are conducted on the datasets provided by HippoRAG 2, which contains 1,000 questions sampled from the validation sets of HotpotQA, 2WikiMultiHopQA, and MuSiQue. The dataset statistics are shown in Table 1.

Dataset	Passages	Queries	Hops
MuSiQue	11,656	1,000	2:518
			3:316
			4:166
2Wiki	6,119	1,000	2:765 4:235
HotpotQA	9,811	1,000	2:1000

Table 1: Dataset Statistics. In the Hops column, $n:m$ indicates that m questions in the dataset require n passages to answer.

Implementation Our implementation is primarily based on PyTorch and the Hugging Face Transformers library, and experiments are conducted on two NVIDIA H200 GPUs. HGRAG is implemented in an iterative manner and leverages sparse matrix operations to enhance computational efficiency. We will make the code publicly available upon the publication of this work.

The main hyperparameters for our method include: the entity similarity vector construction threshold η , the semantic enhancement balancing parameter β , and the diffusion steps t . Best hyperparameter settings for each dataset are shown in Table 2.

B. Hyperparameter Sensitivity Analysis

We analyze the hyperparameter sensitivity of the diffusion steps t on three datasets, and the retrieval results are shown in Figure 1. The overall performance trend is correlated with the difficulty of the datasets. The datasets can be ordered by increasing difficulty as HotpotQA < 2Wiki < MuSiQue, and their overall Recall@5 performance exhibits a corresponding descending trend: HotpotQA > 2Wiki > MuSiQue. For

Dataset	η	β	t
MuSiQue	0.99	6×10^{-4}	4
2Wiki	0.98	1×10^{-4}	3
HotpotQA	0.99	3×10^{-2}	2

Table 2: Hyperparameter Settings.

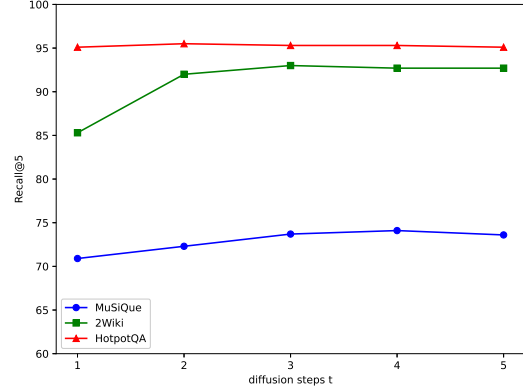


Figure 1: Sensitivity analysis of the hyperparameter t

the simplest dataset, HotpotQA, which requires only 2 reasoning hops, the performance is insensitive to t , and even $t = 1$ also achieves comparable results to higher t . For 2Wiki, which requires 2-4 reasoning hops, $t = 1$ fails to deliver satisfactory performance. A significant improvement is observed at $t = 2$, with the best performance at $t = 3$, and stable results for $t > 1$. For the most challenging dataset MuSiQue, which also requires 2-4 reasoning hops but with a higher average number of hops and a larger proportion of 3/4-hop questions compared to 2Wiki, the best performance is reached at $t = 4$, and performance stabilizes for $t > 2$. These findings suggest the potential for future work to adaptively adjust the value of t based on the difficulty of individual questions.

C. Discussion and Future Work

Our approach offers an alternative perspective for MHQA by using hypergraphs, rather than traditional KGs, to model passage associations. This design significantly reduces graph size and improves retrieval efficiency. While KGs can provide explicit and detailed knowledge, hypergraphs offer a more lightweight and efficient alternative. As LLMs increasingly demonstrate the ability to reason over coarse-grained textual evidence, the trade-off between structural detail and retrieval efficiency becomes an increasingly important design consideration. We adopt the latter approach, and our experimental results validate the effectiveness of this design.

Future work includes extending HGRAG to other knowledge-intensive tasks, integrating it with CoT reasoning methods, and exploring adaptive mechanisms for selecting diffusion steps based on question difficulty.